# From Data Pain to a Data Product

## Scenario

ShopNow e-commerce store

You're supporting:

- BI team: daily KPI dashboard (revenue, orders, active customers)
- ML team: churn model (needs stable daily customer features)

Current pain:

- Dashboard numbers change unexpectedly ("two truths")
- Duplicates appear after retries >> inflated KPIs
- Late events arrive up to 7 days late
- Occasional schema changes (new columns, renames)

Your job: Create a reliable, reproducible Gold product: gold.customer_daily_kpis (1 row per customer per day)

## Input dataset (Bronze)

Bronze table: `bronze.web_events_raw` (append-only, "as-is")

Assumptions:

- Duplicates may exist (`event_id` can repeat due to resend/retry)
- Events may arrive late (up to 7 days)
- Schema may evolve (columns added, sometimes renamed)

Schema:

- `event_id` (string)
- `event_type` (string: `page_view`, `add_to_cart`, `checkout_started`, `purchase`, `refund`)
- `customer_id` (string, nullable for anonymous events)
- `event_time` (timestamp, event time)

- `order_id` (string, nullable)
- `order_total_eur` (decimal, nullable)
- `source` (string: `web`, `ios`, `android`)
- `ingest_time` (timestamp)
- `batch_id` (string)

Sample of `bronze.web_events_raw`:

| event_id | event_type | customer_id | event_time (UTC) | order_id | order_total_eur | source | ingest_time (UTC) |
|---|---|---|---|---|---|---|---|
| e1 | page_view | c1 | 10/02/26 10:01 | null | null | web | 10/02/26 10:01 |
| e2 | checkout_started | c1 | 10/02/26 10:05 | o10 | null | web | 10/02/26 10:05 |
| e3 | purchase | c1 | 10/02/26 10:07 | o10 | 29.95 | web | 10/02/26 10:07 |
| e3 | purchase | c1 | 10/02/26 10:07 | o10 | 29.95 | web | 10/02/26 10:08 |
| e4 | page_view | c2 | 10/02/26 12:10 | null | null | ios | 10/02/26 12:10 |
| e5 | purchase | c2 | 10/02/26 12:20 | o11 | 15.00 | ios | 10/02/26 12:20 |
| e6 | refund | c2 | 11/02/26 09:00 | o11 | 15.00 | ios | 17/02/26 09:01 |
| e7 | purchase | c3 | 10/02/26 23:58 | o12 | 40.00 | android | 11/02/26 00:02 |
| e8 | purchase | c3 | 09/02/26 18:00 | o09 | 10.00 | android | 11/02/26 00:03 |
| e9 | add_to_cart | null | 10/02/26 11:00 | null | null | web | 10/02/26 11:00 |

## Your target

Data product: `gold.customer_daily_kpis`

Consumers:

- BI dashboard (daily KPIs)
- ML churn model (daily customer features)

Design constraints

- Must support backfills (recompute past days)
- Must be idempotent (re-running same interval produces same output)
- Must handle late data up to 7 days
- Must define stable semantics (so metrics don't drift silently)

## Tasks

A. Write two concrete questions the business/ML wants answered.
B. Define at least 2 metrics, and for each metric include: definition + population + window + timezone/late-data rule
C. Declare the gold grain and primary key
D. Design (not implement) the 3-node DAG (Bronze -> Silver -> Gold):
    a. Node 1 (Bronze): what do you store "as-is"? what metadata?
    b. Node 2 (Silver): where/how do you deduplicate and normalize?

   c. Node 3 (Gold): what aggregations? what semantics frozen here?

E. Write the contract (simplified minimum set) for `gold.customer_daily_kpis` including, at least, five schema fields, two constraints (quality rules) and one SLO, following the provided template.

F. Identify 2 breaking-change risks and respective mitigations